

Docker Compose Lab

Flask + Nginx + Custom Network + Multi-stage Build + Non-root User

- ✓ Multi-stage Dockerfile (builder → production)
- ✓ Non-root user (appuser) — never run as root
- ✓ Custom Nginx image with reverse proxy config
- ✓ docker-compose.yml — 2 services, 1 custom bridge network
 - ✓ Container DNS — nginx finds flask by service name
- ✓ Every compose command: up, down, logs, exec, scale, pull

```
ac4e7eb8825b  react-multistage:v1  "/docker-entrypoint...."  2 days ago      Up 2 days      0
b14a9b6e900f  react-multistage:v1  "/docker-entrypoint...."  2 days ago      Up 2 days      0
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes$ ls
data  devboard  postgres.env
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes$ mkdir compose-lab && cd compose-lab
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ mkdir web nginx
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ ls
nginx  web
○ syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ █
```

0 0 5



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS 5
syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ cat > web/app.py << 'EOF'
from flask import Flask
import os

app = Flask(__name__)

<p>Container hostname: <strong>{}</strong></p>
<p>Traffic came through Nginx reverse proxy!</p>
''.format(os.popen('whoami').read().strip(),
          os.popen('hostname').read().strip())

@app.route('/health')
def health():
    return {'status': 'healthy', 'service': 'flask'}

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
EOF
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ cat > web/requirements.txt << 'EOF'
flask==3.0.0
EOF
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ cat web/app.py
cat web/requirements.txt
from flask import Flask
import os

app = Flask(__name__)

@app.route('/')
def home():
    return ''
    <h1>🐳 Docker Compose Lab</h1>
    <p>Flask is running as user: <strong>{}</strong></p>
    <p>Container hostname: <strong>{}</strong></p>
    <p>Traffic came through Nginx reverse proxy!</p>
    ''.format(os.popen('whoami').read().strip(),
              os.popen('hostname').read().strip())

@app.route('/health')
def health():
    return {'status': 'healthy', 'service': 'flask'}

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
flask==3.0.0
○ syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS 5 bash -
syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ cat > web/Dockerfile << 'EOF'
# — Stage 1: Builder —————
FROM python:3.11 AS builder

WORKDIR /app ...
COPY --from=builder /root/.local /home/appuser/.local
COPY --from=builder /app .

RUN chown -R appuser:appuser /app

USER appuser

EXPOSE 5000

CMD ["python", "app.py"]
EOF
• syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ cat web/Dockerfile
# — Stage 1: Builder —————
FROM python:3.11 AS builder

WORKDIR /app

COPY requirements.txt .
RUN pip install --user --no-cache-dir -r requirements.txt

COPY . .

# — Stage 2: Production (slim + non-root) ———
FROM python:3.11-slim AS production

RUN groupadd -r appuser && useradd -r -g appuser appuser

WORKDIR /app

COPY --from=builder /root/.local /home/appuser/.local
COPY --from=builder /app .

RUN chown -R appuser:appuser /app

USER appuser

EXPOSE 5000

CMD ["python", "app.py"]
○ syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$
```

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```
syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ cat > docker-compose.yml << 'EOF'
version: '3.8'
```

```
services:
```

```
...
```

```
networks:
```

```
  app-network:
```

```
    driver: bridge
```

```
EOF
```

```
• syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ cat docker-compose.yml
```

```
version: '3.8'
```

```
services:
```

```
  web:
```

```
    build:
```

```
      context: ./web
```

```
      dockerfile: Dockerfile
```

```
      container_name: flask-app
```

```
      networks:
```

```
        - app-network
```

```
      restart: always
```

```
      environment:
```

```
        - FLASK_ENV=production
```

```
  nginx:
```

```
    build:
```

```
      context: ./nginx
```

```
      dockerfile: Dockerfile
```

```
      container_name: nginx-proxy
```

```
      ports:
```

```
        - "80:80"
```

```
      networks:
```

```
        - app-network
```

```
      depends_on:
```

```
        - web
```

```
      restart: always
```

```
networks:
```

```
  app-network:
```

```
    driver: bridge
```

```
○ syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$
```

```
[+] up 5/5
✓ Image compose-lab-nginx      Built      44.3s
✓ Image compose-lab-web       Built      44.3s
✓ Network compose-lab_app-network Created    0.1s
✓ Container flask-app         Started    0.7s
✓ Container nginx-proxy       Started    1.0s

syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
94a305bb8306	compose-lab-nginx	"/docker-entrypoint..."	20 seconds ago	Up 19 seconds	0.0.0.0:80->80/tcp, [::]:80->80/tcp	nginx-proxy
74fd804e0c34	compose-lab-web	"python app.py"	20 seconds ago	Up 19 seconds	5000/tcp	flask-app
44e532f393c1	nginx	"/docker-entrypoint..."	About an hour ago	Up About an hour	80/tcp	container2
98df6006c585	nginx	"/docker-entrypoint..."	About an hour ago	Up About an hour	80/tcp	container1
ea635a63d49e	nginx	"/docker-entrypoint..."	3 hours ago	Up 3 hours	80/tcp	web2
e59cd82131f5	nginx	"/docker-entrypoint..."	3 hours ago	Up 3 hours	80/tcp	web1
991ebe3b1621	nginx	"/docker-entrypoint..."	3 hours ago	Up 3 hours		isolated-container
816ebd232a43	postgres	"docker-entrypoint.s..."	3 hours ago	Up 3 hours	0.0.0.0:5444->5432/tcp, [::]:5444->5432/tcp	pg-lab-2
4a57c1b56d6d	postgres	"docker-entrypoint.s..."	26 hours ago	Up 26 hours	0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp	mydb
a1cb16ad479c	react-multistage:v1	"/docker-entrypoint..."	26 hours ago	Up 26 hours	0.0.0.0:8083->80/tcp, [::]:8083->80/tcp	My-New-Cont
ac4e7eb8825b	react-multistage:v1	"/docker-entrypoint..."	2 days ago	Up 2 days	0.0.0.0:8082->80/tcp, [::]:8082->80/tcp	volume-demo3
b14a9b6e900f	react-multistage:v1	"/docker-entrypoint..."	2 days ago	Up 2 days	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp	react-app

```
syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$
```

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1:21 AM
6/17/2026



Docker Compose Lab

Flask is running as user: **appuser**

Container hostname: **74fd804e0c34**

Traffic came through Nginx reverse proxy!


```
b14a9b6e900f  react-multistage:v1  "/docker-entrypoint...."  2 days ago      Up 2 days      0.0.0.0:8080->80/tcp, [::]:8080->80/tcp      react-app
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ docker compose ps
WARN[0000] /home/syedaftab04/docker-labs/volumes/compose-lab/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
NAME                IMAGE                COMMAND                SERVICE  CREATED          STATUS              PORTS
flask-app           compose-lab-web      "python app.py"        web      About a minute ago  Up About a minute   5000/tcp
nginx-proxy         compose-lab-nginx    "/docker-entrypoint...."  nginx    About a minute ago  Up About a minute   0.0.0.0:80->80/tcp, [::]:80->80/tcp
○ syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$
```



```

syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ docker compose logs
WARN[0000] /home/syedaftab04/docker-labs/volumes/compose-lab/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
nginx-proxy | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
nginx-proxy | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
nginx-proxy | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
nginx-proxy | 10-listen-on-ipv6-by-default.sh: info: /etc/nginx/conf.d/default.conf is not a file or does not exist
nginx-proxy | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
flask-app    | * Serving Flask app 'app'
flask-app    | * Debug mode: off
flask-app    | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
flask-app    | * Running on all addresses (0.0.0.0)
flask-app    | * Running on http://127.0.0.1:5000
flask-app    | * Running on http://172.20.0.2:5000
flask-app    | Press CTRL+C to quit
flask-app    | 172.20.0.3 - - [17/Jun/2026 05:21:01] "GET / HTTP/1.1" 200 -
flask-app    | 172.20.0.3 - - [17/Jun/2026 05:21:02] "GET /favicon.ico HTTP/1.1" 404 -
nginx-proxy | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
nginx-proxy | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
nginx-proxy | /docker-entrypoint.sh: Configuration complete; ready for start up
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: using the "epoll" event method
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: nginx/1.31.1
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: built by gcc 15.2.0 (Alpine 15.2.0)
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: OS: Linux 6.6.114.1-microsoft-standard-WSL2
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:1048576
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: start worker processes
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: start worker process 20
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: start worker process 21
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: start worker process 22
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: start worker process 23
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: start worker process 24
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: start worker process 25
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: start worker process 26
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: start worker process 27
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: start worker process 28
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: start worker process 29
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: start worker process 30
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: start worker process 31
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: start worker process 32
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: start worker process 33
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: start worker process 34
nginx-proxy | 2026/06/17 05:20:27 [notice] 1#1: start worker process 35
nginx-proxy | 172.20.0.1 - - [17/Jun/2026:05:21:01 +0000] "GET / HTTP/1.1" 200 218 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/149.0.0.0 Safari/537.36" "-"

```

```
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ curl localhost:80/health
{"service":"flask","status":"healthy"}
○ syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ docker compose logs -f web
WARN[0000] /home/syedaftab04/docker-labs/volumes/compose-lab/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please r
onfusion
flask-app | * Serving Flask app 'app'
flask-app | * Debug mode: off
flask-app | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
flask-app | * Running on all addresses (0.0.0.0)
flask-app | * Running on http://127.0.0.1:5000
flask-app | * Running on http://172.20.0.2:5000
flask-app | Press CTRL+C to quit
flask-app | 172.20.0.3 - - [17/Jun/2026 05:21:01] "GET / HTTP/1.1" 200 -
flask-app | 172.20.0.3 - - [17/Jun/2026 05:21:02] "GET /favicon.ico HTTP/1.1" 404 -
flask-app | 172.20.0.3 - - [17/Jun/2026 05:23:10] "GET / HTTP/1.1" 200 -
flask-app | 172.20.0.3 - - [17/Jun/2026 05:23:32] "GET /health HTTP/1.1" 200 -
flask-app | 172.20.0.3 - - [17/Jun/2026 05:24:49] "GET / HTTP/1.1" 200 -
flask-app | 172.20.0.3 - - [17/Jun/2026 05:24:49] "GET / HTTP/1.1" 200 -
flask-app | 172.20.0.3 - - [17/Jun/2026 05:24:50] "GET / HTTP/1.1" 200 -
flask-app | 172.20.0.3 - - [17/Jun/2026 05:24:50] "GET / HTTP/1.1" 200 -
flask-app | 172.20.0.3 - - [17/Jun/2026 05:24:51] "GET / HTTP/1.1" 200 -
|
```

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```
6 (KHTML, like Gecko) Chrome/149.0.0.0 Safari/537.36" "-"
```

```
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ curl localhost:80
```

```
<h1>🐳 Docker Compose Lab</h1>
```

```
<p>Flask is running as user: <strong>appuser</strong></p>
```

```
<p>Container hostname: <strong>74fd804e0c34</strong></p>
```

```
<p>Traffic came through Nginx reverse proxy!</p>
```

```
○ syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$
```

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<p>container: no-name: 400 Bad Request</p>

<p>Traffic came through Nginx reverse proxy!</p>

- syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab\$ curl localhost:80/health
{"service":"flask","status":"healthy"}
- syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab\$


```
flask-app | 172.20.0.3 - - [17/Jun/2026 05:24:50] "GET / HTTP/1.1" 200 -  
flask-app | 172.20.0.3 - - [17/Jun/2026 05:24:51] "GET / HTTP/1.1" 200 -  
docker compose exec web bash^C  
syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ docker compose exec web bash  
WARN[0000] /home/syedaftab04/docker-labs/volumes/compose-lab/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potent  
onfusion  
appuser@74fd804e0c34:/app$ whoami  
appuser  
appuser@74fd804e0c34:/app$
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS 6

```
syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ cat docker-compose.yml
```

```
web:
  build:
    context: ./web
    dockerfile: Dockerfile
  networks:
    - app-network
  restart: always
  environment:
    - FLASK_ENV=production

nginx:
  build:
    context: ./nginx
    dockerfile: Dockerfile
  container_name: nginx-proxy
  ports:
    - "80:80"
  networks:
    - app-network
  depends_on:
    - web
  restart: always

networks:
  app-network:
    driver: bridge
```

```
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ docker compose up -d --scale web=2
```

```
WARN[0000] /home/syedaftab04/docker-labs/volumes/compose-lab/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
```

[+] up 4/4

```
✓ Container compose-lab-web-2 Started 12.4s
✓ Container flask-app Recreated 10.5s
✓ Container nginx-proxy Running 0.0s
✓ Container compose-lab-web-1 Started 0.7s
```

```
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ docker compose ps
```

```
WARN[0000] /home/syedaftab04/docker-labs/volumes/compose-lab/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
compose-lab-web-1	compose-lab-web	"python app.py"	web	23 seconds ago	Up 11 seconds	5000/tcp
compose-lab-web-2	compose-lab-web	"python app.py"	web	23 seconds ago	Up 11 seconds	5000/tcp
nginx-proxy	compose-lab-nginx	"/docker-entrypoint...."	nginx	9 minutes ago	Up 3 minutes	0.0.0.0:80->80/tcp, [::]:80->80/tcp

```
syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$
```

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```
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ docker compose down
WARN[0000] /home/syedaftab04/docker-labs/volumes/compose-lab/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] down 4/4
✓ Container nginx-proxy      Removed    0.4s
✓ Container compose-lab-web-2 Removed    10.5s
✓ Container compose-lab-web-1 Removed    10.4s
✓ Network compose-lab_app-network Removed    0.3s
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ docker compose ps
WARN[0000] /home/syedaftab04/docker-labs/volumes/compose-lab/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
NAME          IMAGE          COMMAND          SERVICE   CREATED   STATUS    PORTS
○ syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$
```

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```
WARN[0000] /home/syedaftab04/docker-labs/volumes/compose-lab/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential c
onfusion
NAME      IMAGE      COMMAND      SERVICE      CREATED      STATUS      PORTS
• syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ docker compose down --rmi all
WARN[0000] /home/syedaftab04/docker-labs/volumes/compose-lab/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential c
onfusion
[+] down 2/2
✓ Image compose-lab-web:latest      Removed      0.1s
✓ Image compose-lab-nginx:latest    Removed      0.1s
• syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$
```

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1:32 AM
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```
✓ Image compose-lab-web:latest Removed
✓ Image compose-lab-nginx:latest Removed
syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ docker images | grep compose-lab
syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
nginx:latest	608a100c7165	241MB	66MB	U
postgres:latest	29ee7bb30d80	650MB	168MB	U
react-multistage:v1	e23246145e60	93.1MB	26.1MB	U
ubuntu:latest	f3d28607ddd7	160MB	45.3MB	U

```
syedaftab04@SYEDBASIT26:~/docker-labs/volumes/compose-lab$
```

0 0 5

